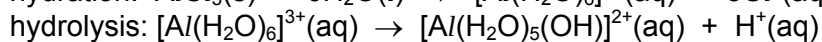
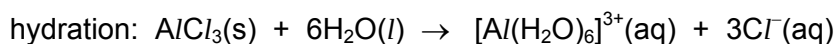


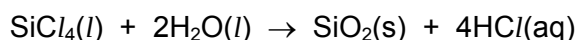
Suggested Solutions for 2011 JC2 Preliminary Examination Paper 3

- 1 (a) (i) $K_w = [\text{H}^+][\text{OH}^-]$
- (ii) This is because the self-ionisation process of water is endothermic. According to Le Chatelier's Principle, increase in temperature favours endothermic process. At higher temperature, equilibrium position in $\text{H}_2\text{O}(l) = \text{H}^+(\text{aq}) + \text{OH}^-(\text{aq})$ shifts right to remove some heat and thus results in higher $[\text{H}^+]$ (i.e. lower pH).
- (iii) At 40°C , $\text{p}K_w = \text{pH} + \text{pOH} = 2 \times 6.7 = 13.4$
 $[\text{OH}^-] = \sqrt{c \times K_b} = \sqrt{0.100 \times 1.86 \times 10^{-5}} = 1.36 \times 10^{-3} \text{ mol dm}^{-3}$
 $\text{pH} = \text{p}K_w - \text{pOH} = 13.4 - (-\lg 1.36 \times 10^{-3}) = 10.5$

- (b) (i) AlCl_3 hydrolyses in water to form weakly acidic solution of pH 3. Due to high charge density of Al^{3+} , Al^{3+} polarises the electron cloud of the H_2O molecules, thereby weakens and breaks the O–H bond in the H_2O molecule to release H^+ ion.

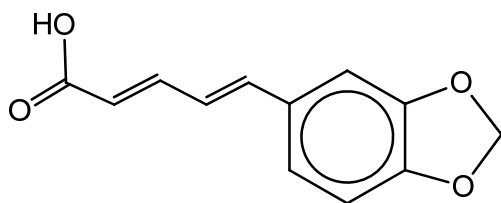


SiCl_4 hydrolyses in water to give a strongly acidic solution of pH 2 because Si has energetically accessible vacant 3d orbitals.

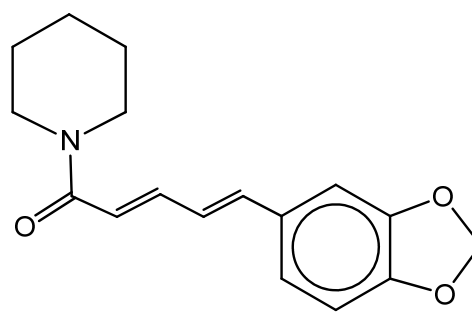


- (ii) $\text{Al}_2\text{O}_3(\text{s}) + \text{Na}_2\text{O}(\text{s}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{Na}[\text{Al}(\text{OH})_4](\text{aq})$
- (c) (i) Br_2 is reduced since O.S. of Br changes from 0 in Br_2 to -1 in HBr .
 SO_2 is oxidised since O.S. of S changes from $+4$ in SO_2 to $+6$ in SO_3 .
- (ii) $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$
- (iii) The colour of the resulting mixture changes from colourless to light brown/yellow upon addition of first drop of excess Br_2 .

- (d) (i)



A



Piperine

- (ii) Step I: LiAlH_4 , dry ether
 Step II: acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, warm with immediate distillation
- (iii) Piperonylic acid is a stronger acid than piperonyl alcohol.

The negative charge on $-\text{COO}^-$ is delocalised over 2 O atoms which disperses the negative charge on the anion and thus stabilising the anion.

- 2 (a) (i) Lithium electrode: $\text{Li} \rightarrow \text{Li}^+ + \text{e}^-$
Carbon electrode: $2\text{Li}^+ + 2\text{e}^- + \text{O}_2 \rightarrow \text{Li}_2\text{O}_2$
- (ii) $E^\circ_{\text{cell}} = E^\circ(\text{O}_2/\text{Li}_2\text{O}_2) - E^\circ(\text{Li}^+/\text{Li})$
 $E^\circ(\text{O}_2/\text{Li}_2\text{O}_2) = E^\circ_{\text{cell}} + E^\circ(\text{Li}^+/\text{Li}) = +3.1 + (-3.04) = +0.06 \text{ V}$
- (iii) Amt of $\text{Li}_2\text{O}_2 = 37.8 / 45.8 = 0.825 \text{ mol}$
 Since $\text{Li}_2\text{O}_2 \equiv 2\text{e}^-$, amt of e^- , $n = 2 \times 0.825 = 1.65 \text{ mol}$
 Using $Q = It = nF$, $t = nF/I = (1.65)(96500)/10.0 = 15844 \text{ s} = 4.42 \text{ h}$ OR 4.40 h
- (a) (iv) Lithium electrode (cathode): negative
Carbon electrode (anode): positive
- (v) Any one of the following:
- Lithium is very reactive. It burns explosively with oxygen or water.
 - Lithium is a fire hazardous material which will further ignite in times of car accidents/overheated.
 - Lithium peroxide is a very dangerous fire hazardous material as it is a strong oxidizing agent.
 - Lithium peroxide gives off a lot of heat (or highly exothermic) with reacted with water or steam.
- (b) (i) $\text{Li}_2\text{O}_2 \equiv \text{CO}_2$
 Amt of Li_2O_2 needed = Amt of CO_2
 $= 1000 / (12.0 + 32.0)$ OR $1000 / 44.0$
 $= 22.7 \text{ mol}$
 Mass of $\text{Li}_2\text{O}_2 = 22.7 \times 45.8 = 1040 \text{ g}$ OR 1.04 kg
- (ii) Since $2\text{LiOH} + \text{CO}_2 \rightarrow \text{Li}_2\text{CO}_3 + \text{H}_2\text{O}$,
 Li_2O_2 offers the bonus of releasing/supplying oxygen.
- (iii) Since Group I metal ions have a lower charge and bigger size than Group II metal ions, Group I metal ions have a lower charge density than Group II metal ions. Thus, Group I metal ions will polarise the large electron cloud of CO_3^{2-} to a smaller extent. Hence Group I metal carbonates are more thermally stable and require a higher decomposition temperature.
- (c) (i)
- $\text{Li}^+ \left[\begin{array}{c} \text{H} \\ \bullet \times \\ \text{H} \times \text{Al} \times \text{H} \\ \bullet \times \\ \text{H} \end{array} \right]^-$
- (ii)
-
- (iii) Step I : limited $\text{Cl}_2(\text{g})$, uv light
Step II : $\text{Cl}_2(\text{aq})$
- (iv) Step III: Acid-base reaction
Step IV: Nucleophilic Substitution

- 3 (a) (i) From solution formed from halogen A:

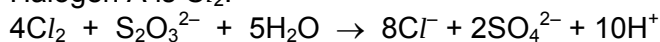
White precipitate, C, formed with AgNO_3 is AgCl .

White precipitate, D, formed with $\text{Ba}(\text{NO}_3)_2$ is BaSO_4 .

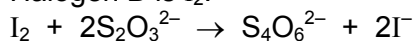
From solution formed from halogen B:

Yellow precipitate, E, formed with AgNO_3 is AgI .

- (ii) Halogen A is Cl_2 .



Halogen B is I_2 .



- (iii) From the Data Booklet: $E^\circ(\text{Cl}_2/\text{Cl}^-) = +1.36 \text{ V}$; $E^\circ(\text{I}_2/\text{I}^-) = +0.54 \text{ V}$

Since $E^\circ(\text{Cl}_2/\text{Cl}^-)$ is more positive than $E^\circ(\text{I}_2/\text{I}^-)$, Cl_2 is a stronger oxidising agent than I_2 .

- (b) (i) $\bullet\text{OH}(\text{g}) + \text{HCl}(\text{g}) \rightarrow \bullet\text{Cl}(\text{g}) + \text{H}_2\text{O}(\text{l})$

- (ii) Bond energy is the amount of energy required to break one mole of H-Cl covalent bonds in gaseous molecules into its constituent H and Cl gaseous atoms.

- (iii)
$$\begin{aligned}\Delta H_{\text{rxn}} &= E(\text{O-H}) + E(\text{H-Cl}) - 2E(\text{O-H}) \\ &= E(\text{H-Cl}) - E(\text{O-H}) \\ &= 431 - 460 \\ &= -29 \text{ kJ mol}^{-1}\end{aligned}$$

- (iv) From the table, % decomposition of HX increases down the group. The thermal stability of HX decreases down the group.

From the *Data Booklet*,

$$E(\text{H-Cl}) = 431 \text{ kJ mol}^{-1}; E(\text{H-Br}) = 366 \text{ kJ mol}^{-1}; E(\text{H-I}) = 299 \text{ kJ mol}^{-1}$$

This is because down the group, the H-X bond energy decreases and H-X bond is weaker.

OR

From the *Data Booklet*:

Atomic radius of $\text{Cl} = 0.099 \text{ nm}$;

Atomic radius of $\text{Br} = 0.114 \text{ nm}$

Atomic radius of $\text{I} = 0.133 \text{ nm}$

This is because down the group, the atomic radius of X becomes larger and thus the H-X bond becomes longer and weaker.

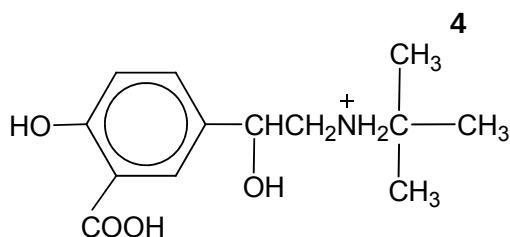
- (c) (i) Test: Add $\text{Br}_2(\text{aq})$ to both samples separately.
Observations: Salbutamol decolourises brown Br_2 but not ephedrine.

OR

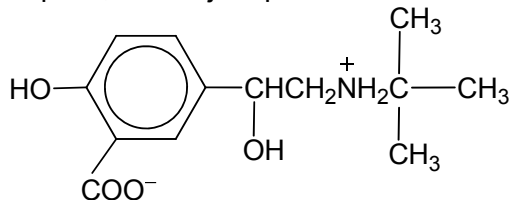
Test: Add neutral $\text{FeCl}_3(\text{aq})$ to both samples separately.

Observations: Salbutamol will give a violet colouration but not ephedrine.

- 3 (c) (ii) At pH 4, the major species is a cation which migrates towards the cathode.



At pH 8, the major species is a zwitterion and hence no migration is observed.



- (iii) Salbutamol has a lower melting point than compound **G**.

Both are simple molecular structures with hydrogen bonds between the molecules. However, for Salbutamol, it is possible to form intramolecular hydrogen bonds between the phenol group and the primary alcohol group. Thus, less energy is required to overcome the less extensive intermolecular hydrogen bonds between salbutamol molecules as compared to that in **G**.

- 4 (a) (i) $x = 0.0150 \text{ mol dm}^{-3} \text{ s}^{-1}$
 $y = 0.0300 \text{ mol dm}^{-3} \text{ s}^{-1}$
- (ii) $\text{Rate} = k_2 [\text{Q}][\text{OH}^-]$
 $\therefore k_2 = \frac{\text{Rate}}{[\text{Q}][\text{OH}^-]} = \frac{0.0240}{0.30 \times 0.40} = 0.200 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$
- (iii) From $\text{Rate} = k_2 [\text{Q}][\text{OH}^-]$
 $\Rightarrow \text{Rate} = k'' [\text{Q}]$ where $k'' = k_2 [\text{OH}^-]$
 and $t_{1/2}$ of **Q** = $\ln 2/k'' = z$ seconds in expt 4

Comparing expt 4 & 5

In expt 5, $[\text{OH}^-]$ is increased to $3/2$ times of original value in expt 4,
 k'' is increased to $3/2$ times and thus $t_{1/2}$ of **Q** is decreased to $2/3$ of original value.
 $\therefore$ Half-life of **Q** in expt 5 = $z \times 0.4/0.6 = 2/3 z$ seconds

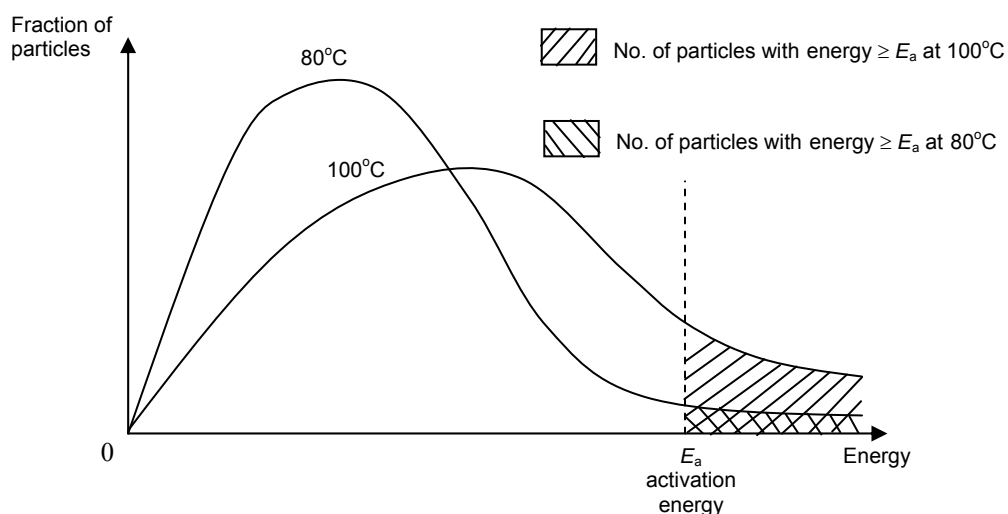
Comparing expt 4 & 6

In expt 6, $[\text{OH}^-]$ is half of that in expt 4, k'' is halved and thus $t_{1/2}$ of **Q** is doubled.
 $\therefore$ Half-life of **Q** in expt 6 = $2z$ seconds
 (Although $[\text{Q}]$ is doubled in expt 6, it has no effect on the half-life of **Q** as half-life of **Q** is independent of $[\text{Q}]$ since rxn is first order with respect to **Q**.)

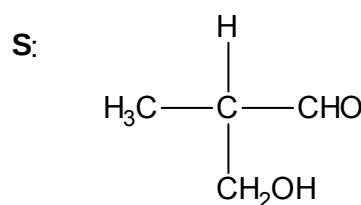
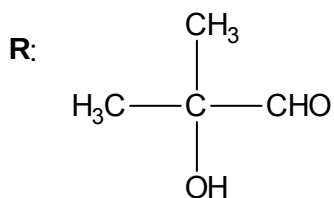
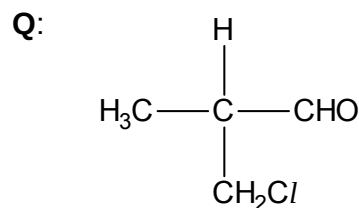
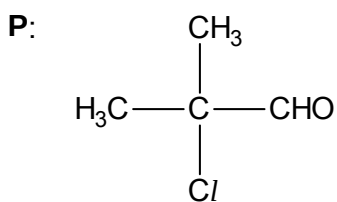
- 4 (b) (i) When the temperature of the reaction increases (pressure remaining the same),

the reactant particles gain more kinetic energy and hence move about more rapidly.

Number of particles having energy $\geq E_a$ increases. Frequency of effective collision between reactant particles increases and thus the rate of reaction increases.

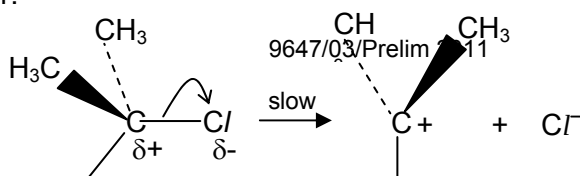


- (ii) Rate of reaction remains the same as the reactants are in the liquid state/ not gaseous and are not affected by pressure (as liquids are not compressible).
- (c) (i)
- Compounds P & Q are branched $\rightarrow$ P and Q are either primary or tertiary alkyl halides
 - Compounds P & Q reduced Fehling's solution $\rightarrow$ P and Q are aldehydes
 - P and Q, both alkyl halides, undergo nucleophilic substitution with NaOH to form R and S, both alcohols, respectively.
 - Since the reaction is first order with respect to P but zero order with respect to NaOH, the reaction is an S_N1 reaction $\rightarrow$ P is a tertiary alkyl halide
 - Since the reaction is first order with respect to Q and with respect to NaOH, the reaction is an S_N2 reaction $\rightarrow$ Q is a primary alkyl halide



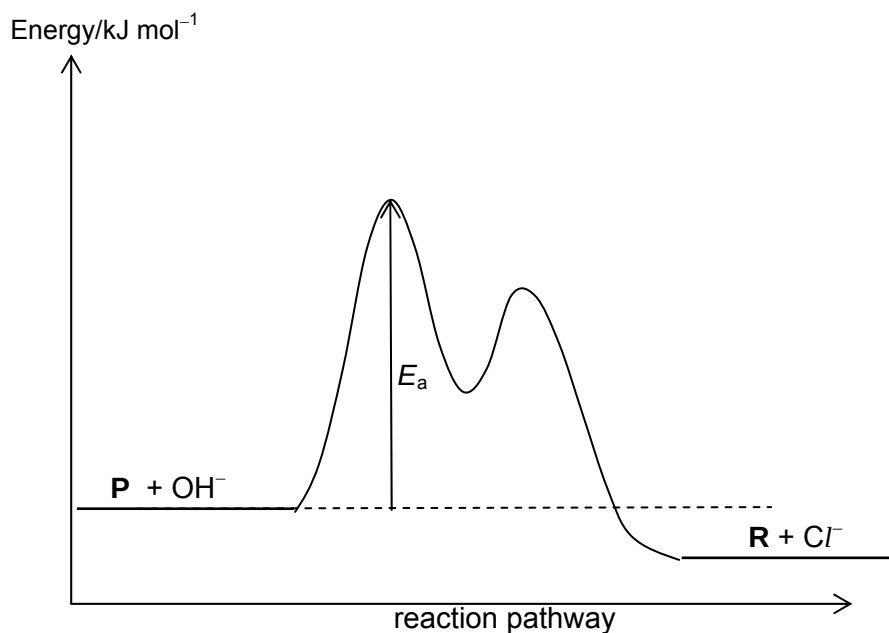
4 (c) (ii) Step 1:

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Step 2:

(iii)



- (d) (i) Rate of hydrolysis of butanoyl chloride is faster than 2-chlorobutane to produce Cl^- so that white ppt of $AgCl$ is precipitated out immediately.

In the acyl chloride, the carbon in $>C=O$ is highly electron-deficient as it is bonded to two electronegative atoms, O & Cl. Hence, this carbon is more susceptible to attack by the OH^- nucleophile as compared with 2-chlorobutane.

OR

Less steric hindrance at the carbonyl carbon as it is sp^2 hybridised.

- 4 (d) (ii) Rate of formation of ppt: $CH_3CHICH_2CH_3 > CH_3CHClCH_2CH_3 > CH_2=CClCH_2CH_3$

From *Data Booklet*, $E(\text{C}-\text{Cl}) = 340 \text{ kJ mol}^{-1}$; $E(\text{C}-\text{I}) = 240 \text{ kJ mol}^{-1}$

C-I bond is weaker than C-Cl bond due to smaller bond energy

OR

C-I bond is longer and weaker due to larger I atom.

Hence C-I bond is easier to break.

C-Cl bond in $\text{CH}_2=\text{CClCH}_2\text{CH}_3$ is stronger due to overlapping of p-orbital of Cl with the π electron cloud of the C=C bond, resulting in partial double bond character.

OR

Presence of C=C bond adjacent to the C-Cl makes the carbon atom of C-Cl less electron deficient and hence less susceptible to nucleophilic attack.

5	(a)	(i)	$2\text{SO}_2(\text{g})$	+	$\text{O}_2(\text{g})$	=	$2\text{SO}_3(\text{g})$
			Initial partial pressure / atm	2	1		–
			Eqm partial pressure / atm	$2 - 2(1 \times \frac{95}{100})$ = 0.100	$1 - (1 \times \frac{95}{100})$ = 0.05		$2(1 \times \frac{95}{100})$ = 1.90

$$K_p = \frac{(1.90)^2}{(0.100)^2(0.05)} = 7220 \text{ atm}^{-1}$$

(ii) Equilibrium position will not change since the rate of both the forward and backward reactions are increased to the same extent.

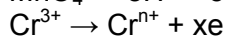
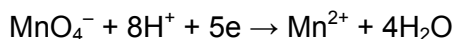
(iii) Heterogeneous catalysis.

Vanadium is able to have variable oxidation states because variable number of 3d and 4s electrons can be involved in bonding. since 3d and 4s electrons have similar energies.

OR

Vanadium has partially filled 3d and 4s orbitals for adsorption of reactants to occur. It allows the reactants to be adsorbed onto the surface, increasing surface concentrations of the reactants, thus increasing the rate of reaction.

(b) (i) The green coloured ion is $\text{Cr}(\text{H}_2\text{O})_6^{3+}/\text{Cr}^{3+}$.



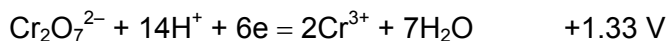
Amount of electrons = $5 \times 0.020 = 0.100 \text{ mol}$

Ratio of blue ion : e = $0.10 : 0.100 = 1:1$

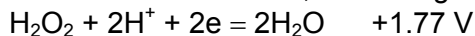
The blue coloured ion is $\text{Cr}^{2+}/\text{Cr}(\text{H}_2\text{O})_6^{2+}$.

5 (b) (ii) Since question stated that Cr^{2+} can only be oxidised to Cr^{3+} using KMnO_4 , a stronger

oxidising agent than KMnO_4 is needed to oxidise Cr^{3+} to $\text{Cr}_2\text{O}_7^{2-}$.



From the *Data Booklet*, the stronger oxidising agents include:

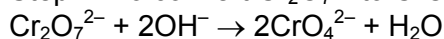


F_2 will react with water rather than Cr^{3+} , and $\text{S}_2\text{O}_8^{2-}$ is unstable at room temperature. Hence H_2O_2 is the best option.

Step 1: Add H_2O_2 :

$$E_{\text{cell}} = +1.77 - (+1.33) = +0.44 \text{ V} > 0 \text{ (energetically feasible)}$$

Step 2: To convert $\text{Cr}_2\text{O}_7^{2-}$ to CrO_4^{2-} , add aqueous NaOH .



(c)

